Name:- Rushi babasaheb Naikawade

Branch:- (AI&DS)

Roll No. :- B-13

Batch :- SYB-B1

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AIM :Implement and analyze various probability distributions in Python, including Binomial, Poisson, and Normal distribution
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THEORY : 1 . Difference between Binomial, Poisson, and Normal Distributions ? ANS. a.Binomial Distribution Type: Discrete Parameters: n (number of trials), p (probability of success) Trials: Fixed number of trials Outcome: Each trial has two possible outcomes — success or failure Shape: Symmetric when p = 0.5, otherwise can be skewed Explanation: The Binomial Distribution models the number of successes in a fixed number of independent trials, where each trial has the same probability of success. Example: Tossing a coin multiple times and counting the number of heads b.Poisson Distribution Type: Discrete Parameter: λ (lambda — average rate of occurrence) Trials: Not fixed; events occur over a continuous interval (time, space, etc.) Outcome: Counts the number of events in a given interval Shape: Right-skewed when λ is small; becomes more symmetric as λ increases Explanation: The Poisson Distribution models how many times an event occurs in a fixed interval, assuming events happen independently and at a constant average rate. Example: Number of phone calls received in an hour c.Normal Distribution Type: Continuous Parameters: μ (mean), σ (standard deviation) Trials: Not based on trials Outcome: Can take any real value Shape: Symmetric, bell-shaped curve Explanation: The Normal Distribution describes continuous data that clusters around a mean. It is widely used because many natural phenomena follow this pattern. Example: Heights of people

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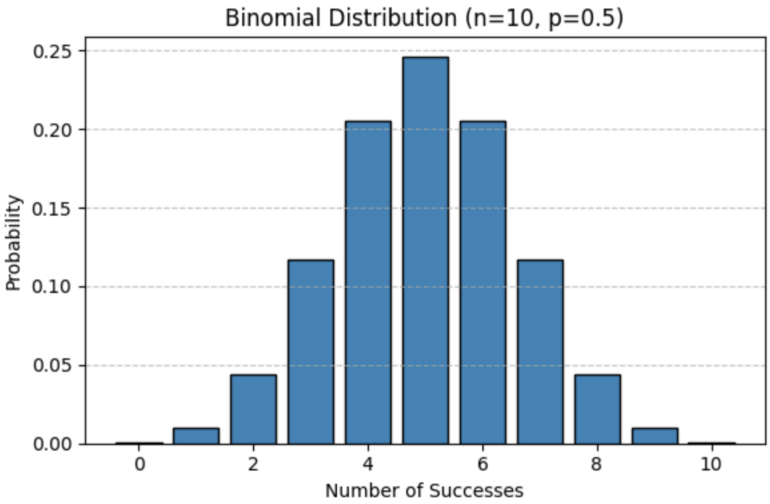
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import binom, poisson, norm
import warnings
warnings.filterwarnings('ignore') # Corrected typo here

# ---------------------------------------------#
# 1. BINOMIAL DISTRIBUTION
# ---------------------------------------------
n = 10 # number of trials
p = 0.5 # probability of success
x_binom = np.arange(0, n+1)
binom_pmf = binom.pmf(x_binom, n, p)
plt.figure(figsize=(6, 4))
plt.bar(x_binom, binom_pmf, color='steelblue', edgecolor='black')
plt.title('Binomial Distribution (n=10, p=0.5)')
plt.xlabel('Number of Successes')
plt.ylabel('Probability')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()

# Mean and Variance
print("=== Binomial Distribution ===")
print(f"Mean     : {binom.mean(n, p)}")
print(f"Variance : {binom.var(n, p)}")
print(f"Skewness : {binom.stats(n, p, moments='s')}")
```




```
=== Binomial Distribution ===
Mean     : 5.0
```